
Buy versus Rent in England: A Cash-Flow-Matched Horse Race, 2005–2026

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Abstract

I compare the financial outcomes of buying a first home with renting and investing using a monthly cash-flow-matched simulation on official English data from January 2005 to June 2026. The buyer bears transaction costs, mortgage interest, maintenance and other user costs of homeownership, while the counterfactual renter occupies the same dwelling at market rent and invests the deposit, purchase costs and every subsequent monthly saving in a global equity index fund. Over the full sample period, renting produces 44% greater final wealth. Across 199 five-year starting cohorts, however, the two strategies are remarkably close: buying wins 51% of cohorts, with a median wealth advantage of only £994. Buying wins more often at longer horizons—62% of seven-year cohorts and 61% of ten-year cohorts—although median renter wealth remains within 8–11% of owner wealth. Outcomes depend strongly on timing and leverage: an unlevered cash buyer wins only 8% of cohorts, while regional buy-win rates vary substantially across England. Finally, I derive a required appreciation rate that translates the historical comparison into a forward-looking break-even condition. Under June 2026 conditions, a five-year buyer requires house-price growth of 2.3% per year to match a renter earning a cautious 5.9% annual equity return.

JEL classification: D14, G11, G51, R21, R31

Keywords: housing tenure choice; buy versus rent; first-time buyers; user cost of housing; household portfolios; leverage; stamp duty

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1 Introduction

Buying a home is among the largest financial decisions most households will ever make. It also carries significant cultural weight: homeownership is widely regarded as an important life milestone and is often assumed to be financially superior to renting. Around 65% of English households currently own their home (Ministry of Housing, Communities and Local Government, 2025), while the difficulty faced by many younger households in getting onto the property ladder has become a prominent concern. Yet renting is not necessarily just “dead money”. Once the full costs of homeownership are accounted for, together with the opportunity cost of capital tied up in a property, the financial advantage of owning becomes less clear. If a renter invests the cash that would otherwise be committed to a deposit, purchase costs, and higher ongoing expenses, both renting and owning can become financially viable ways of paying for housing.

This paper compares these two strategies for a typical first-time buyer using official English data from January 2005 to June 2026, through a monthly cash-flow-matched simulation following Beracha and Johnson (2012). In the baseline scenario, the buyer makes a 10% deposit and takes out a 30-year mortgage, initially fixed for five years and subsequently refinanced as the loan amortises, with calibrated maintenance, depreciation, transaction costs, and period-accurate stamp duty. The renter occupies the same dwelling at market rent and invests the deposit and purchase costs, together with every subsequent monthly difference in housing outlays, in a global equity index fund (MSCI ACWI, in sterling). I then compare the homeowner’s net equity after selling costs with the value of the renter’s investment portfolio.

Over the full sample period (start 2005, end 2026), renting produces substantially greater final wealth, leaving the renter 44% wealthier than the owner. Across starting cohorts, however, the comparison is much closer. At a five-year horizon—approximately the typical tenure of a first-time buyer (Santander UK, 2024)—buying produces greater terminal wealth in 102 of 199 cohorts (51%). The median renter-to-owner wealth ratio is 0.98, and the median difference is just £994 in the buyer’s favour. Buying becomes more successful at longer horizons, winning 62% of seven-year cohorts and 61% of ten-year cohorts, but even then the median renter finishes within 8–11% of the owner. At ten years, for example, the median wealth gap is £14,454 on a dwelling worth £319,488. The central result is therefore not that one tenure consistently dominates the other, but that renting and buying have been financially much more comparable than the conventional wisdom would suggest.

Which side of this distribution a household falls on depends heavily on when it buys. Renting wins almost every cohort beginning between 2005 and 2010, as well as every cohort beginning in 2016, while buying wins most cohorts beginning in 2011–2015 and again in 2017–2020. Initial housing valuations and financing conditions are particularly informative: the price-to-rent ratio and mortgage rate at purchase jointly account for 56% of the cross-cohort variation. Outcomes also vary geographically, with regional buy-win rates over five-year holds ranging from 38% in the North East to 63% in the West Midlands.

Leverage is central to understanding why buying remains competitive despite much weaker underlying asset returns. Global equities returned 10.4% per year in sterling over the sample, compared with house-price growth of 3.2% per year. A mortgage, however, allows the buyer to earn housing gains on an asset substantially larger than their initial equity contribution, turning relatively modest house-price appreciation into a competitive return on the buyer's own capital. Consistent with this mechanism, an unlevered cash buyer wins only 8% of cohorts, while the win rate peaks at 54% with a 15–20% deposit. The same leverage also magnifies the cost of poor timing. Eleven North East cohorts and five North West cohorts entering between mid-2007 and mid-2008 finish five years later with *negative* net worth after selling costs.

Holding period matters as well, although the pattern should be interpreted as a feature of this particular historical sample rather than as a general property of tenure. At short horizons, the main obstacle is the cost of entering and exiting homeownership: round-trip transaction costs amount to roughly thirty per cent of the buyer's initial outlay and take around six years to amortise. Results at longer horizons, however, increasingly reflect the particular sequence contained in the data—one housing crash, a prolonged period of cheap credit, and an exceptional global equity bull market. For this reason, I do not extend the historical cohort comparison beyond ten years.

Finally, I translate the historical comparison into a forward-looking break-even measure: the required appreciation rate (RAR), defined as the annual rate of house-price growth at which buying produces exactly the same terminal wealth as renting and investing. It is the user-cost condition of Poterba (1984) and Himmelberg, Mayer and Sinai (2005), solved for the capital-gains term. Under June 2026 conditions and a five-year horizon, the dwelling must appreciate by 2.3% per year for buying to match a renter earning a cautious CAPE-implied equity return of 5.9%, but by 3.2% per year if equities instead earn the 10.4% annual return realised over the historical sample.

All data, code, and an interactive implementation of the framework are openly available, and the analysis is designed to be re-run as new months of official data arrive.

2 Related literature

The paper draws on, and speaks to, five strands of work.

User cost and tenure choice. The theoretical benchmark for rent-versus-own comparisons is the user cost of owner-occupied housing (foregone interest on equity, taxes, mortgage interest, maintenance and depreciation, less expected capital gains), formalised by Poterba (1984) on tenure-choice foundations laid by Henderson and Ioannides (1983), and turned into the standard housing-valuation tool by Himmelberg, Mayer and Sinai (2005). A measurement branch shows, however, that ex-ante user costs diverge persistently from observed rents and are acutely sensitive to imputed appreciation and financing assumptions (Verbrugge, 2008; Díaz and Luengo-Prado, 2008), and that credible price-rent comparisons require quality- and composition-matched data

(Hill and Syed, 2016; Bracke, 2015). This paper is the ex-post counterpart to that ex-ante tradition: rather than imputing an equilibrium user cost, I track the realised cash flows of both tenures month by month, which sidesteps the imputation problem; my required appreciation rate is the user-cost equilibrium condition solved for the capital-gains term rather than assuming it; and the composition-consistent national aggregate of Section 3.2 responds directly to the matched-measurement critique.

Buy-versus-rent horse races. The closest methodological ancestors are the ex-post backtests for the United States: Beracha and Johnson (2012), who run cash-flow-matched buy-versus-rent races over 1978–2009 and find renting frequently competitive; the companion indifference-appreciation exercise of Beracha, Seiler and Johnson (2012), an antecedent of my RAR metric computed under stylised ex-ante assumptions rather than realised paths; and Rappaport (2010) and Goodman and Mayer (2018) on homeownership as wealth-building, with Herbert, McCue and Sanchez-Moyano (2013) surveying the distributional evidence. For Britain, the only comparable academic exercise is Mostafa and Jones (2019), an annual-frequency comparison of first-time-buyer cohorts over 1975–2011 in which the renter’s counterfactual earns building-society deposit rates; buying wins essentially always. Tabner (2016) provides a UK net-present-value framework under assumed parameters, and practitioner analyses exist for Canada (Felix and Arif, 2025). Relative to this work, the present paper is the first monthly, cash-flow-matched backtest on UK data, and the first in which the British renter’s counterfactual is a diversified equity portfolio; that single change overturns the always-buy conclusion and produces near-parity, while extending the sample through the post-2012 boom, the pandemic, and the 2022–2023 rate shock that earlier UK samples miss.

Housing returns. A recent measurement literature quantifies long-run returns to residential property: house prices track incomes rather than compounding like equities over the long run (Knoll, Schularick and Steger, 2017); total returns are competitive with equities only once rental yield is included and before costs (Jordà et al., 2019); and micro-level studies show that transaction costs, management and idiosyncratic risk substantially erode net returns to individual owners (Chambers, Spaenjers and Steiner, 2021; Eichholtz et al., 2021; Giacoletti, 2021; Demers and Eisfeldt, 2022), with large spatial heterogeneity (Amaral et al., 2025). My results are the household-level, England-specific counterpart: with realistic costs the unlevered housing return loses badly to equities, since an all-cash buyer wins only 8% of cohorts, and it is mortgage debt, not the housing return itself, that makes ownership financially competitive.

Housing in household portfolios. A structural literature examines how owner-occupied housing shapes household portfolios and lifecycle choices (Flavin and Yamashita, 2002; Cocco, 2005; Yao and Zhang, 2005; Chetty, Sandor and Szeidl, 2017; Vestman, 2019); J. Y. Campbell (2006) and Gomes, Haliassos and Ramadorai (2021) survey the field. These models emphasise precisely the two forces my backtest measures ex post: the portfolio distortion of a large, levered, indivisible housing position, and the financing channel through which mortgage debt amplifies returns to home equity. The empirical leverage gradient I document is, in effect, a realised-

data trace of the mechanism these models study, with the qualification that it is not monotonic: buying wins 46% of cohorts at a 5% deposit, 54% at 15–20%, and only 8% in cash.

Rent risk, commitment, and the UK market. Two channels are deliberately switched off in my accounting. The first is *rent risk*: a tenant’s exposure to future increases in the market rent, which they must pay or move, whereas an owner converts most of the same housing cost into a mortgage payment fixed in nominal terms for the length of the fix. Ownership therefore acts as a hedge against rent risk (Sinai and Souleles, 2005), and my renter is given none of that protection, bearing actual market rents throughout, which over this sample rose 2.9% per year. Mortgage amortisation also acts as forced saving for present-biased households (Laibson, 1997; Thaler and Benartzi, 2004; Schlafmann, 2021), with Bernstein and Koudijs (2024) showing amortisation causally builds net worth and Sodini et al. (2023) identifying substantial benefits of ownership quasi-experimentally, whereas my renter is endowed with perfect investment discipline. Near-parity for a perfectly disciplined renter therefore implies an advantage for ownership among typical households, which is how I read the result. Finally, the paper connects to UK housing economics, covering cycles and regional dynamics (Muellbauer and Murphy, 1997; Meen, 1999), rate-driven valuations (Miles and Monro, 2021), supply constraints (Hilber and Vermeulen, 2016), and the stamp-duty responses documented by Besley, Meads and Surico (2014) and Best and Kleven (2018), which motivate my period-accurate SDLT module. It connects too to the evidence that price-rent ratios forecast housing returns (Case and Shiller, 1989; J. Y. Campbell and Shiller, 1988; Gallin, 2008; S. D. Campbell et al., 2009; Engsted and Pedersen, 2015).

3 Data

The analysis draws on official statistics for all housing inputs and on index-provider data for financial returns. All series are monthly, in nominal pounds sterling, and cover January 2005 to June 2026 (258 months). I focus on England and its nine regions: England is the only part of the United Kingdom for which official monthly rent statistics reach back to 2005.

The sample is bounded at both ends by data rather than by choice. The start is set by rents: no regional private-rent series for England exists at monthly or quarterly frequency before January 2005, in any source, official or commercial. The end is set by prices: the UK House Price Index runs about two months in arrears, so June 2026 is the latest complete vintage even though rents are published to July. Appendix A provides construction details; Table 1 summarises the sources.

Table 1: Data sources

Series	Source
House prices	UK House Price Index, HM Land Registry (all dwellings, England and regions)
Private rents	ONS Price Index of Private Rents (PIPR) and its historical series
Mortgage rate	Bank of England quoted rates: five-year fix at 90% loan-to-value, constructed (Section 3.3)
Equity returns	MSCI ACWI net total return, GBP, less 0.12% p.a. fund fee
Gilt yield	10-year UK government bond yield (OECD, via FRED)
Margin rate	Bank of England Bank Rate plus 1.50% (renter-leverage robustness)
Stamp duty	HMRC SDLT schedules, first-time-buyer rules by month

3.1 House prices and rents

House prices come from the UK House Price Index (UK HPI), the official transaction-based index constructed from HM Land Registry sales records by hedonic regression with geometric averaging (HM Land Registry, 2026). I use the average price of all dwellings, monthly, for England and each region.

Rents come from the ONS Price Index of Private Rents (PIPR), the official measure of private rents based on Valuation Office Agency lettings data (Office for National Statistics, 2026a; Office for National Statistics, 2026b). Two features make PIPR the appropriate counterpart to the UK HPI. First, it publishes average rent *levels* in pounds per calendar month, not only an index. Second, it is a *stock* measure covering new and existing tenancies, matching the position of a long-term renter whose rent adjusts with the market rather than only at new lets. The live PIPR series begins in January 2015; earlier years use the ONS historical series, chain-linked to PIPR’s predecessor index back to January 2005 for England and its regions (Office for National Statistics, 2025c). The two sources agree exactly over their 122-month overlap, so the splice introduces no discontinuity.

3.2 A composition-consistent representative dwelling

Moving from regions to a single national figure raises a significant measurement issue. The headline UK HPI England price and the headline PIPR England rent each average the country, but they weight the regions differently: the price is a transaction-mix average, close to a simple average of the nine regions, whereas the rent is a tenancy-mix average dominated by South East and London’s large private rental sector. In June 2026 the national average rent (£1,446) exceeded the average rent of every region except London, while the national average price (£293,262) sat near the middle of the regional range. The ratio of the two, a price-to-rent ratio of 16.9, consequently falls *below the ratio of every individual region* (17.7 to 22.4). Any rent-versus-own comparison built directly on this pair prices a mid-England home against a more expensive London-weighted rent, overstating the cost of renting and biasing the comparison towards ownership.

I therefore construct the price and rent of a *representative English dwelling* as the same population-weighted average of the nine regions:

$$P_t^{\text{rep}} = \sum_{i=1}^9 w_i P_{i,t}, \quad R_t^{\text{rep}} = \sum_{i=1}^9 w_i R_{i,t}, \quad w_i = \frac{\text{pop}_i}{\sum_j \text{pop}_j}, \quad (1)$$

with fixed weights from mid-2023 ONS regional population estimates. Both aggregates then share one regional composition, and the resulting price-to-rent ratio (20.7 in June 2026) lies within the regional range. Over the sample the representative dwelling appreciated from £162,312 to £319,488 (3.2% per year), falling 18% peak-to-trough during 2007–2009, while its rent rose from £692 to £1,286 per month (2.9% per year). The regional analyses in Sections 5 and 6 use each region’s own price and rent and are unaffected by aggregation.

3.3 Mortgage rates

The rate a first-time buyer pays is not the rate the average borrower pays. English first-time buyers overwhelmingly fix: of new first-time-buyer lending in the first half of 2025, 99.1% was fixed-rate and 0.018% originated on a reversion rate, and among the fixed loans the three-to-five-year band was the modal choice in six of the seven half-years the Financial Conduct Authority publishes (Financial Conduct Authority, 2025). The baseline borrower therefore takes a **five-year fix at 90% loan-to-value**, refixed at the prevailing five-year rate every sixty months.

No single Bank of England series measures that: the five-year 90% loan-to-value series begins in February 2019. Earlier months are constructed by interpolating in loan-to-value between the published five-year series at 75% and 95%, with a sixty-one-month gap during 2008–2013, when fewer than three lenders offered a 95% five-year fix, filled from the two-year 90% series. Appendix A gives the construction and its validation. The series averages 4.74% and ranges from 2.33% to 6.99%.

Two features of it matter for the results. It sits 1.19 percentage points above the Bank’s effective rate on all new advances (CFMBJ95). That series is volume-weighted across every loan-to-value tier and fixation length, so recasting it monthly removes the fixation cliff through which a British first-time buyer is actually exposed to rate moves; Section 7 reports it as a bounding case. And because lenders price in bands and a borrower moves between them, the rate is repriced at each refix by the borrower’s *current* loan-to-value. A buyer entering at 90% on a thirty-year term is at 74.8% by their first refix, and charging them the 90% premium then, a premium that peaked at 2.2 percentage points over 2010–2014, would overstate the cost of owning by several percentage points.

3.4 Financial returns

The renter’s portfolio tracks the MSCI ACWI index, covering large- and mid-capitalisation equities in 47 developed and emerging markets (MSCI Inc., 2026). I use end-of-month index levels

computed by MSCI directly in pounds sterling in the net-total-return variant (dividends reinvested after withholding taxes), less an ongoing charge of 0.12% per year representing a low-cost global index fund available to UK retail investors. Measuring the index in sterling embeds the currency exposure an unhedged UK investor bears. Over the sample the net-of-fee series returned 10.4% per year, a factor of 8.38. As a home-bias robustness check I also use a FTSE 100 total-return series, which returned 7.1% per year, a factor of 4.38; the choice between the two turns out to be significant, and Section 7 reports it in full. The 10-year gilt yield (OECD, via FRED) serves as a conservative alternative opportunity cost in Section 6; it ranged from 5.4% (June 2007) to 0.2% (July 2020) and stood at 4.9% in May 2026, its latest observation (Federal Reserve Bank of St. Louis, 2026).

Figure 1 shows the model's housing inputs together with the price-to-rent ratio they imply; Figure 2 compares cumulative asset performance. The gap the buyer must overcome is evident: over twenty-one years the global equity portfolio multiplied roughly eightfold, UK large-cap equities fourfold, and the representative English dwelling, before the owner's running costs, twofold.

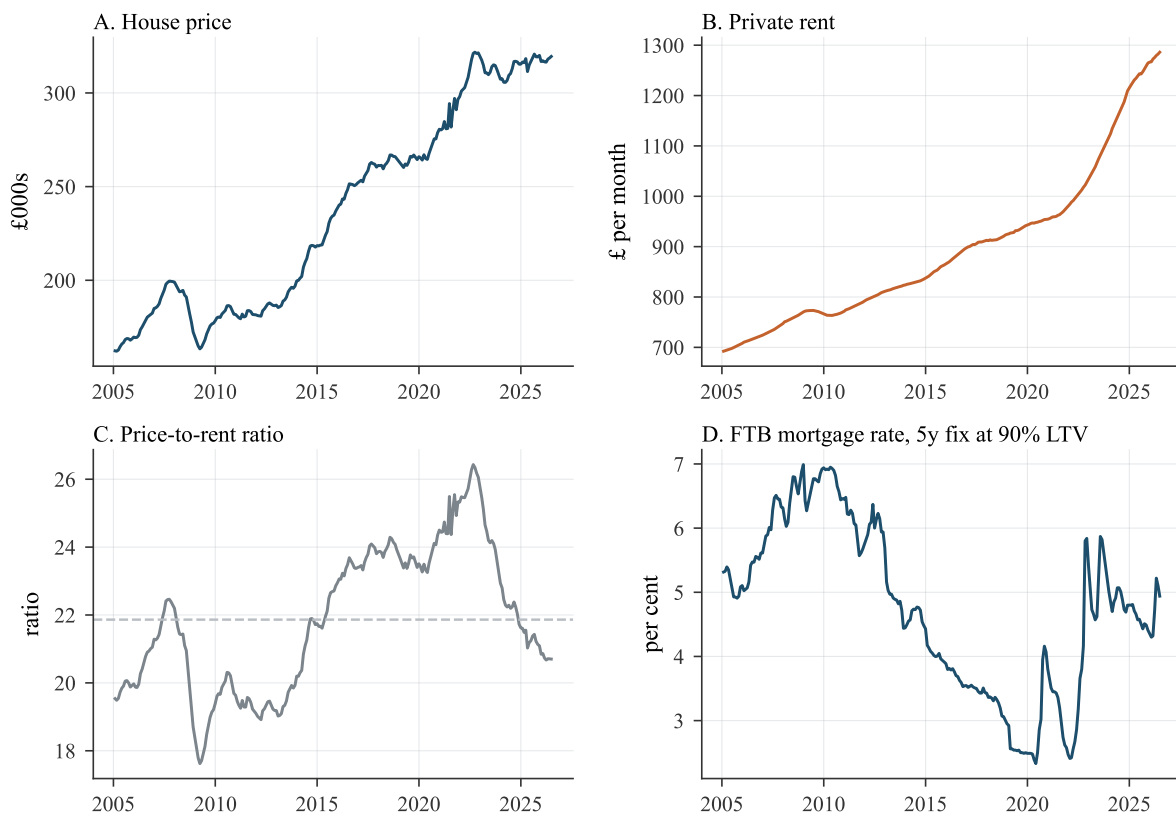


Figure 1: Housing-market inputs for the representative English dwelling, 2005–2026. Panel A: average house price. Panel B: average private rent. Panel C: price-to-rent ratio (price divided by annual rent); the dashed line marks the sample mean of 21.9. Panel D: Bank of England effective interest rate on new mortgage advances to households. Sources: HM Land Registry UK HPI; ONS PIPR; Bank of England.

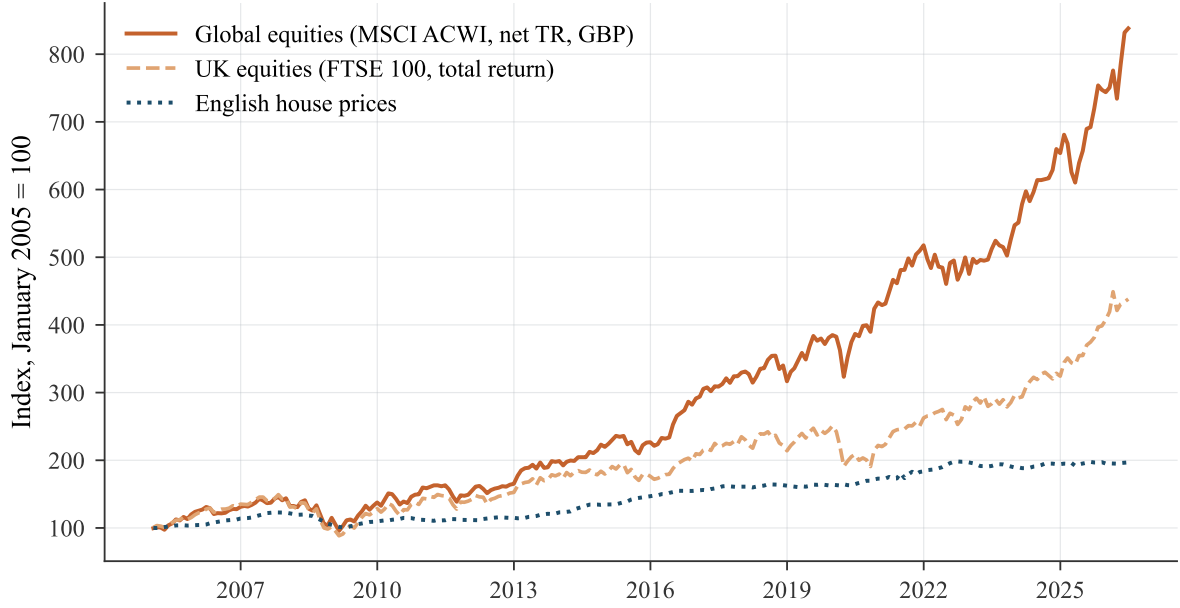


Figure 2: Cumulative performance of global equities (MSCI ACWI, net total return in GBP, less a 0.12% fund fee), UK equities (FTSE 100 total return) and representative English house prices, January 2005 = 100. Sources: MSCI; Morningstar; HM Land Registry.

4 Empirical framework

The simulation compares two housing tenure strategies on a cash-flow-matched basis: both start with identical capital, and their cash flows are equalised every month. Whenever the renter pays less for housing in a given month, the difference is invested; whenever the renter pays more, the difference is withdrawn from the portfolio. The design follows Beracha and Johnson (2012), with transaction costs and taxes calibrated to the English first-time buyer.

4.1 The buyer

The buyer purchases the property at the market price prevailing in the entry month. In the baseline scenario, they put down a deposit of 10% of the price (loan-to-value of 90%), pay purchase costs of 1.2% of the price (conveyancing, survey, mortgage and valuation fees), and stamp duty under the first-time-buyer schedule in force in that month. The mortgage amortises over 30 years.

The baseline rate is a five-year fix, because 99% of first-time-buyer lending is fixed-rate and the five-year band is the modal choice. The payment is set at entry from the quoted five-year rate at the buyer's loan-to-value, held constant for 60 months, then reset at the rate prevailing at the refix:

$$r_{m,t} = (1 + r_t)^{1/12} - 1, \quad \text{Payment}_t = \frac{B_t r_{m,t}}{1 - (1 + r_{m,t})^{-(T-t)}}, \quad (2)$$

where r_t is the annual rate in force, B_t the outstanding balance and $T - t$ the remaining months. At each refix the rate is taken from the band matching the loan-to-value the buyer has amort-

ised down to, so a borrower who started at 90% is not charged the high-loan-to-value premium indefinitely. Appendix A sets out the rate construction. A floating rate and a two-year fix are examined in Section 7.

Each month the owner also pays maintenance and depreciation of 1.5% per year of the property's current market value,

$$M_t = \frac{0.015}{12} V_t, \quad (3)$$

an all-in charge covering routine maintenance, major works and replacements, buildings insurance, and physical depreciation not offset by spending. It is one of the most consequential assumption in the paper, so it is built from the national accounts rather than from convention.

The charge has two components. *Depreciation* is measured by ONS consumption of fixed capital on dwellings, divided by the market value of the dwelling stock including the land beneath it: 1.20% per year on average over 1995–2024, 1.28% in 2024, with a range of 0.95% to 1.47% (Office for National Statistics, 2025a; Office for National Statistics, 2025d). *Cash maintenance and buildings insurance* add roughly 0.35%, from ONS Family Spending (Office for National Statistics, 2025b). Together, 1.55%, which the baseline rounds to 1.5%.

The land share is what makes this figure specific to the United Kingdom, and it is why figures imported from the American literature do not directly transfer. Depreciation applies to structures, not to the ground they stand on, and in the UK land is 69.9% of the market value of household residential property. A rate estimated on structure value, or on total value in a market where land is half of it, therefore overstates the charge here by a wide margin once applied to a British price. On a UK basis, ONS implies depreciation of 4.24% of *structure* value, which is 1.28% of market value at a 30.1% structure share.

Section 7 reports 1.2% to 1.8% as the range UK evidence supports, the floor being depreciation alone, for a buyer who does no discretionary upkeep, and the ceiling being the peak observed depreciation year plus cash maintenance. Values of 1% and 2–3% are also reported, but as bounds beyond the evidence rather than as equally supported alternatives.

The property value itself evolves with the relevant house-price index, $V_{t+1} = V_t(1 + g_t)$. The buyer's terminal wealth after a holding period ending at T^* is net housing equity after selling costs of 1.8% (estate-agent commission, conveyancing, energy certificate):

$$W^{\text{own}} = V_{T^*}(1 - 0.018) - B_{T^*}. \quad (4)$$

4.2 The renter

The renter occupies an equivalent dwelling at the market rent prevailing each month and invests all spare cash. The initial portfolio equals the buyer's total initial outlay (deposit, purchase costs and stamp duty), and each month the difference between the owner's outflow (mortgage payment plus maintenance) and the rent is added to, or withdrawn from, the portfolio, which earns the monthly return on the global equity index (net of withholding taxes and fund fees). The renter's

terminal wealth is the portfolio value, $W^{\text{rent}} = P_{T^*}$.

4.3 Outcome measure and analyses

Outcomes are summarised by the renter/owner wealth ratio

$$R = \frac{W^{\text{rent}}}{W^{\text{own}}}, \quad (5)$$

$R > 1$ means renting-and-investing produced greater terminal wealth. The ratio is invariant to the initial capital scale and comparable across cohorts and regions.

Four analyses are run: (i) 199 entry cohorts, one for every month from January 2005 to July 2021, each holding for exactly five years, which is the primary result; (ii) the same exercise repeated at every holding period from one to ten years; (iii) both region by region, aggregated with population weights; and (iv) the forward-looking required appreciation rate of Section 6, at end-of-sample conditions.

Cohorts start every month which should not be mistaken for extra evidence: the underlying sample is the same 258 months either way, and it contains only 4.3 non-overlapping five-year windows however often the simulation is restarted. Cohort counts in this paper are counts of overlapping simulations, not of independent observations.

4.4 Stamp duty

Stamp Duty Land Tax is modelled period-accurately: each simulated purchase pays the tax implied by the England first-time-buyer schedule in force in that month, including the 2008–2009 holiday, the 2010–2012 first-time-buyer relief, the 2014 slab-to-slice reform, the permanent relief from November 2017, the 2020–2021 holiday, and the 2022–2025 threshold changes (HM Revenue & Customs, 2026; House of Commons Library, 2025). Appendix B tabulates the schedules. For the representative dwelling, a first-time buyer paid SDLT in 143 of 258 purchase months (55%), averaging £1,825 (0.88% of the price) when payable and never exceeding £3,144. Stamp duty is therefore a second-order factor nationally. It is not second-order in London, where the average dwelling now sits above the £500,000 ceiling at which first-time-buyer relief is withdrawn entirely, and the bill jumps to £17,694.

4.5 Scope and abstractions

The model excludes several factors by design. Income and capital-gains taxes are omitted on both sides: a first-time buyer’s main residence is exempt from capital-gains tax, and the renter’s flows fit within annual ISA allowances for most of the sample, so the symmetric no-tax treatment is a reasonable approximation for the typical household (though not for very large portfolios held outside tax wrappers). Council tax and utilities are paid by the occupant under both tenure options and net out. In the baseline the renter never borrows to invest, so leverage is available only through the mortgage; Section 7 relaxes that and lets the renter borrow on margin. Non-financial

considerations, among them security of tenure, freedom to modify, flexibility to move and the insurance value of ownership against local rent risk (Sinai and Souleles, 2005), are outside the accounting and are discussed separately in Section 8. Table 2 collects the baseline parameters.

Table 2: Baseline parameters (first-time buyer)

Parameter	Description	Value
Property	Representative English dwelling (Section 3.2)	—
Holding period	Typical tenure in a first home	5 years
Deposit	Share of purchase price (90% loan-to-value)	10%
Mortgage term	Amortisation period	30 years
Rate type	Five-year fix, refixed at the prevailing five-year rate, repriced by current loan-to-value	—
Purchase costs	Conveyancing, survey, mortgage product fee	1.2%
Stamp duty	England first-time-buyer schedule, by month	rule-based
Selling costs	Estate agent, conveyancing, EPC	1.8%
Maintenance & depreciation	Share of current market value, per year	1.5%
Equity index	MSCI ACWI net total return, GBP, less 0.12% fee	—
Renter's leverage	Unlevered portfolio	1.0×
Rent	Market rent, updated monthly	—

5 Results

5.1 The full-sample race

A buyer and a renter both start in January 2005 and are evaluated in June 2026. The buyer purchases the representative dwelling for £162,312 with a deposit of £16,231 and purchase costs of £3,571 including £1,623 of stamp duty; the renter starts with the identical £19,802 in the equity index and occupies the same dwelling at £692 per month. Figure 3 shows the two wealth paths.

The renter finishes 44% ahead, with £367,821 against the owner's £254,658 (a £319,488 dwelling less £59,079 of outstanding mortgage and £5,751 of selling costs). Figure 4 shows the matched flows. The owner's outlay exceeded the rent in 188 of 258 months, so the renter added to the portfolio in most of them: the mortgage payment starts at £801 against rent of £692, and maintenance grows from £203 to £399 per month as the property appreciates. Because the rate refixes every five years rather than floating, the payment moves in steps and peaks at £820 instead of tracking the 2009 collapse downwards; by June 2026 it is £687 against rent of £1,286.

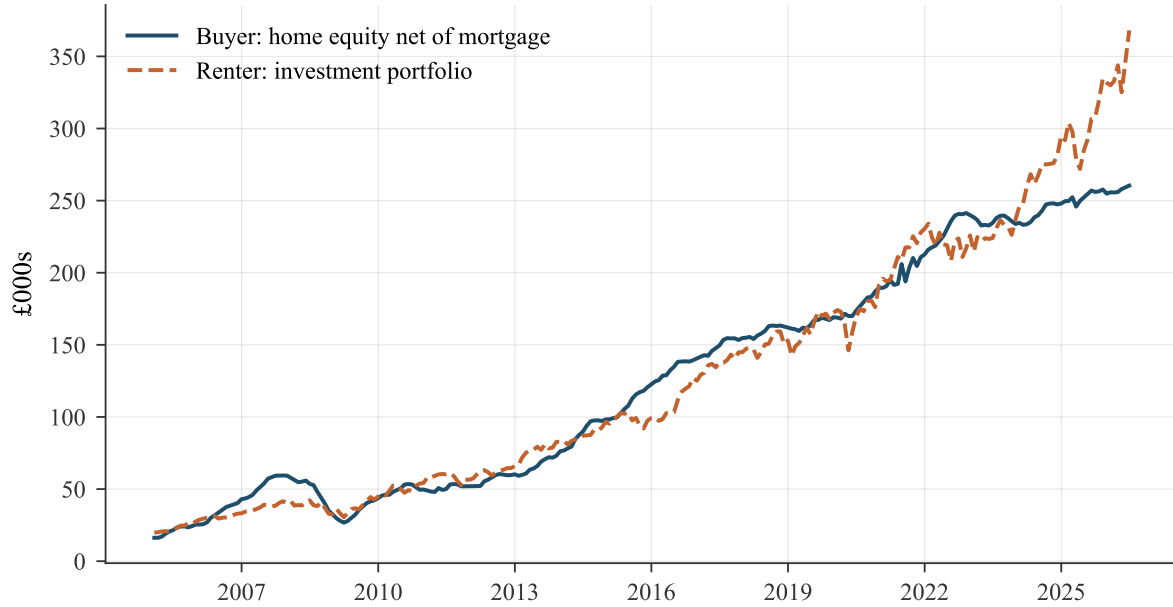


Figure 3: Terminal wealth of the two strategies, January 2005 entry cohort held to June 2026. The buyer's wealth is the market value of the dwelling less the outstanding mortgage; the renter's is the value of the investment portfolio. Selling costs are applied at the terminal date only.

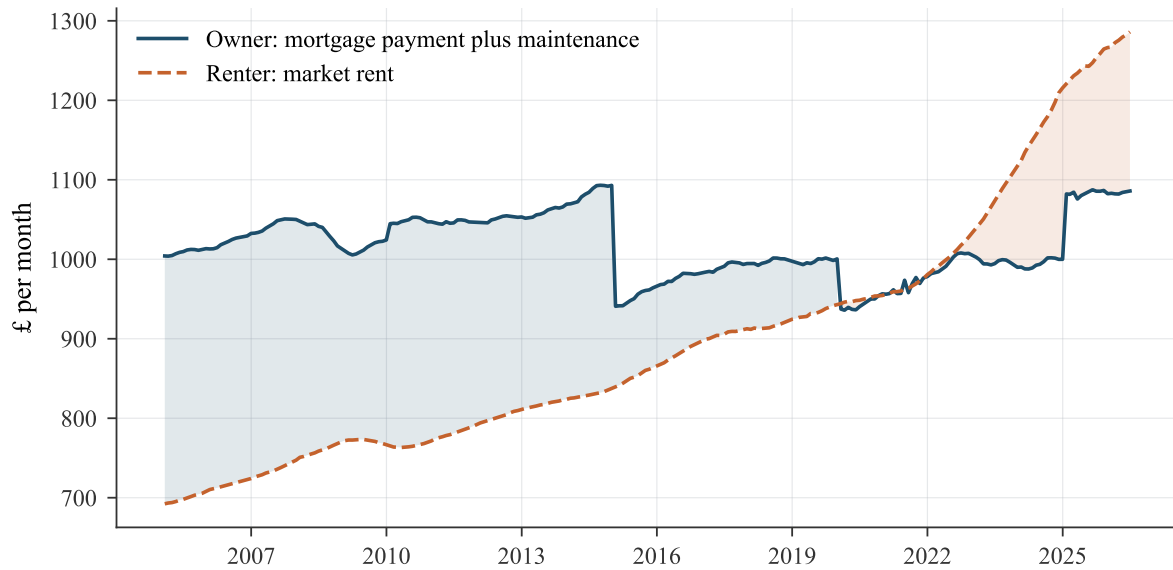


Figure 4: Monthly housing outlays of the two strategies, January 2005 entry cohort. The owner pays the mortgage payment plus maintenance; the renter pays the market rent. Shading marks the gap that is invested (owner pays more) or withdrawn (renter pays more) each month under cash-flow matching.

5.2 Entry cohorts over a five-year hold

The single 2005 start is one draw from history. To isolate timing, I repeat the race for every entry month, each cohort holding for exactly five years.

Five years is chosen because it is roughly how long a first-time buyer keeps their first home: Santander put the average at 4.5 years (Santander UK, 2024). Fixing the holding period also

makes the accounting coherent. Every cohort faces the same round trip (stamp duty and 1.2% purchase costs on the way in, 1.8% selling costs on the way out), and the selling costs are charged at the moment the buyer actually incurs them, when they move on.

This gives 199 cohorts entering between January 2005 and July 2021. Buying produced greater wealth in 102 of them (51%) and renting in 97 (49%). The median wealth ratio is $R = 0.980$ and the median wealth gap £994 in the buyer's favour. On the central case the two strategies finish level. The cohorts overlap heavily, so these are counts of simulations rather than of independent histories: 258 months of data contain only 4.3 non-overlapping five-year windows.

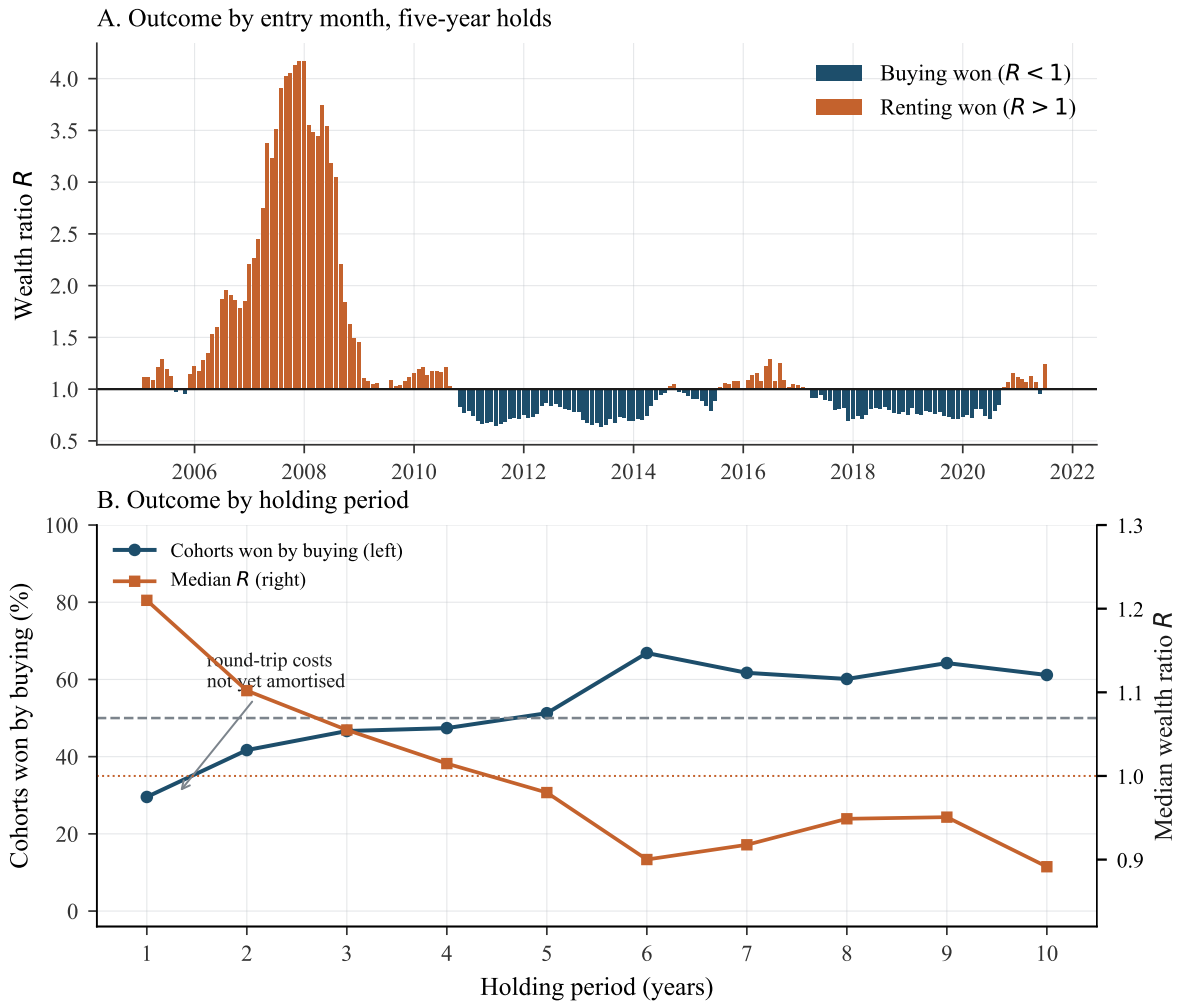


Figure 5: Cohort outcomes. Panel A: renter/owner wealth ratio R by entry month, 199 cohorts each holding for five years. Bars extend from unity; values above one indicate that renting-and-investing produced greater terminal wealth. Panel B: the share of cohorts won by buying at each holding period from one to ten years (left axis), with the median wealth ratio at each horizon on the right axis; monthly entry months at every horizon. Baseline parameters of Table 2.

The tails are lopsided. Renting finished more than twice as wealthy in 21 cohorts, more than three times in 16 of those and more than four times in five; the five all entered between August and December 2007, and the sixteen between April 2007 and July 2008. The peak is the

November 2007 cohort at $R = 4.165$, where the owner finished five years with £15,769 against the renter's £65,685, a gap of £49,916. Buying's best cohort is May 2013 at $R = 0.639$, with the owner on £100,336 against the renter's £64,123, a gap of £36,214.

In terms of timing, buying won 17% of cohorts entering in 2005 and none at all of those entering in 2006, 2007, 2008 or 2009. Its share then rises through 33% in 2010 to every cohort entering in 2011, 2012 and 2013, and 83% of those entering in 2014. The run breaks twice: 2015 split evenly, and renting won every cohort entering in 2016, whose five years ran through the equity gains of 2019–2021. Buying then took 92% of 2017 entries, every cohort entering in 2018 and 2019, 67% in 2020, and one of the seven 2021 cohorts the five-year panel can evaluate, whose holds contain the 2022–2023 rate shock. Stamp duty is a minor factor: 120 of the 199 cohorts paid any, never exceeding 1.2% of the purchase price.

Other holding periods

Panel B of Figure 5 repeats the exercise at every holding period from one to ten years. Buying wins 30% of one-year cohorts, 47% at four years, 51% at five and 60–67% from six to ten. The median moves far less than that share does: the median ratio is 0.918 at seven years and 0.891 at ten, leaving the median renter within 8% and 11% of the owner, and the median ten-year gap is £14,454 on a dwelling worth £319,488. The mean ratio is 1.031 at seven years and 0.963 at ten. The minimum owner net worth is negative at one and two years (–£12,346 and –£8,822).

The short end of the profile is determined by transaction costs. A buyer pays stamp duty plus 1.2% on entry and 1.8% on exit, roughly three per cent of the purchase price in round-trip costs against a 10% deposit, so the transaction itself consumes about thirty per cent of the initial outlay. Over one year there is almost no appreciation or amortisation to set against that, and the buyer loses on arithmetic alone. Each additional year spreads the same fixed cost more thinly, which is why the curve rises steeply to about year six.

This profile is a property of the sample rather than a general fact about tenure. Twenty-one years of English data contain one house-price crash, one long expansion of cheap credit and one exceptional run in global equities, and the shape of the curve depends on where those three episodes fall inside a given holding period.

5.3 Regional results

Running the race region by region, each with its own price *and* rent so that each is internally consistent, validates the representative-dwelling aggregate. Table 3 reports five-year cohort outcomes and the forward break-even appreciation for each region.

Table 3: Regional outcomes over five-year holds, 2005–2026

Region	Price (2026:6)	P/R	Buy- win %	Median <i>R</i>	Median gap	Realised appr.
North East	£165,550	17.7	38	1.11	+£6,309	2.13%
North West	£219,922	19.1	55	0.77	−£7,127	3.26%
London	£553,870	20.1	43	1.12	+£16,223	3.81%
Yorkshire and The Humber	£207,948	20.1	52	0.93	−£2,923	2.93%
West Midlands	£234,731	20.1	63	0.81	−£12,249	3.00%
South West	£304,562	20.5	48	1.01	+£701	2.77%
East Midlands	£240,457	21.8	56	0.95	−£3,327	2.90%
East of England	£338,707	22.0	49	1.01	+£1,089	3.17%
South East	£380,380	22.4	42	1.08	+£6,575	3.12%
England (pop.-weighted)	–	20.6	49	0.98	+£1,263	3.12%

Notes: 199 cohorts per region, each entering in a distinct month and holding for five years. P/R is the June 2026 ratio of price to annual rent. Buy-win % is the share in which buying produced greater terminal wealth. Median *R* is the median renter/owner wealth ratio ($R > 1$: renting won); median gap is the median of renter minus owner terminal wealth, so a negative figure favours the buyer. Realised appr. is the region's own annualised house-price growth over 2005–2026.

The cross-section tracks affordability only loosely. The West Midlands, the East Midlands and the North West (price-to-rent ratios of 20.1, 21.8 and 19.1) delivered the highest buy-win shares at 63%, 56% and 55%, and the largest median gaps in the buyer's favour. The South East, on the highest ratio at 22.4, managed 42%. But the North East breaks the pattern outright: it has the cheapest housing relative to rent of any region, at 17.7, and the *lowest* buy-win share at 38%.

Splitting cohorts by entry era shows where the North East diverges. In the post-crisis window of 2010–2015, when cheap debt made buying win almost everywhere, North East buyers won just 14% of cohorts against 82% in the West Midlands; from 2016 onwards both won essentially every cohort, the North East 98% and the West Midlands all of them. That runs against the national record, where renting won every 2016 entry: the representative dwelling is weighted towards the South East and London, where those cohorts lost. North East prices simply did not recover after 2009, appreciating 2.13% per year across the sample against 3.81% in London. A low price-to-rent ratio makes owning cheap relative to renting, but it cannot substitute for capital growth once transaction costs have to be recovered.

The North East also shows what leverage does when appreciation fails. Eleven of its 199 cohorts, every one of them entering between June 2007 and June 2008, ended a five-year hold with *negative* net worth after selling costs, and five North West cohorts, entering between June 2007 and February 2008, did the same. No other region produced a single instance. A 10% deposit in a low-appreciation region is a position an ordinary downturn can destroy outright.

5.4 Entry conditions

I regress the five-year cohort wealth ratio on the entry mortgage rate and the entry price-to-rent ratio. Jointly the two variables account for 56% of the cross-cohort variation. Pearson r and its R^2 above are bivariate while the t -statistics are on the joint regression, which is why the price-to-rent ratio shows a t of 3.04 despite a bivariate r of 0.03 and p of 0.70: the two regressors are collinear, and the ratio contributes through the joint fit rather than on its own.

	Pearson r	R^2	HAC t
Mortgage rate	0.42	0.18	3.42
Price-to-rent ratio	0.03	0.00	3.04
Joint (OLS)	–	0.56	–

Notes: 199 cohorts, each holding five years. HAC t -statistics use Hansen–Hodrick standard errors with the truncation set to the overlap length (59 months); the corresponding OLS statistics are 15.9 and 13.2.

Through the arithmetic Section 6 makes explicit, these two variables largely *determine* the outcome without forecasting it: buy when the price-to-rent ratio is low and financing is cheap, and the cash flows work in your favour because you have bought cheaply relative to renting.

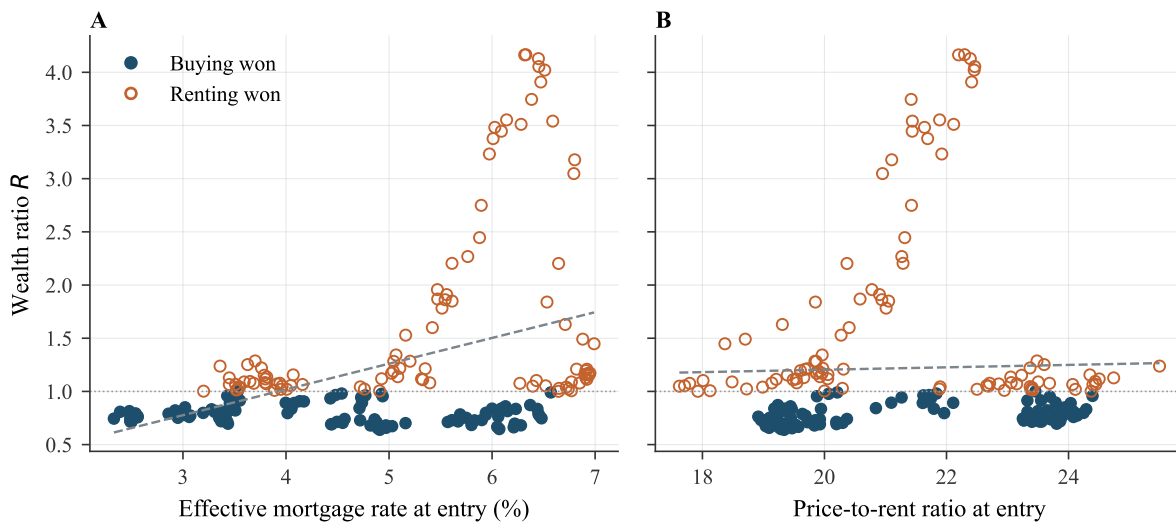


Figure 6: Conditions at entry and the eventual outcome, 199 cohorts each holding five years. Panel A: quoted first-time-buyer mortgage rate at entry. Panel B: price-to-rent ratio at entry. Filled markers denote cohorts in which buying produced greater terminal wealth, hollow markers those in which renting-and-investing did. Dashed lines are ordinary least-squares fits.

6 The required appreciation rate

The cohort analysis of Section 5 looks backwards. It tells a buyer what would have happened had they bought at some past date, which is not the question they face. This section turns the same model forwards.

Define the *required appreciation rate* (RAR) as the constant annual house-price growth at which buying exactly matches renting-and-investing over a given holding period, with every other input held at its assumed path. It is the user-cost equilibrium of Poterba (1984) and Himmelberg, Mayer and Sinai (2005) solved for the one term those authors have to assume. Rather than positing an expected capital gain and pronouncing prices fair or unfair, the I invert the question: reporting the growth rate buying needs, and leaving the reader to supply their own expectation. If prices grow faster than the RAR the buyer wins; slower, and the renter wins.

The RAR is a threshold, so it requires no view on house prices. The one judgement it does need is the return the renter's portfolio earns instead.

6.1 Assumptions

The calculation uses the last month of the sample, June 2026: a representative English dwelling priced at £319,488 let at £1,286 per month, a price-to-rent ratio of 20.7 and a gross rental yield of 4.83%. First-time-buyer stamp duty on that price is £974.

Every parameter of the baseline in Table 2 carries over unchanged: a 10% deposit, a 30-year term, maintenance and depreciation at 1.5% of value per year, purchase costs of 1.2% and selling costs of 1.8%.

The mortgage rate is 4.94%, the June 2026 quoted rate on a five-year fix at 90% loan-to-value (the same series the historical analysis uses, at its latest observation). It is held constant across the horizon. This is a real assumption and it cuts against the buyer at long horizons: a borrower refixing every five years is exposed to rates that may be higher or lower, and holding today's rate fixed for ten years understates that uncertainty in both directions. Rent is assumed to grow at 3% per year, against the 2.9% the representative dwelling's rent actually delivered over the sample.

The renter's return is the input that matters most. The cautious figure is 5.9% nominal: a CAPE-implied long-run real return on global equities of roughly 3%, plus expected inflation of roughly 3%, less the 0.12% fund fee. The realised figure is 10.4%, the return the renter's portfolio actually earned over 2005–2026 net of that fee. The 4.94% ten-year gilt yield provides a third, lower-risk comparator.

6.2 Results

Table 4: Forward-looking required appreciation rate, June 2026 conditions

Horizon	vs. global equities		vs. gilts (4.9%)
	at 5.9% (cautious)	at 10.4% (realised)	
3 years	2.77%	3.53%	2.62%
5 years	2.27%	3.24%	2.08%
7 years	2.03%	3.22%	1.80%
10 years	1.79%	3.38%	1.50%

Notes: Constant annual house-price growth at which buying exactly matches renting-and-investing, at June 2026 inputs. For comparison, the representative English dwelling appreciated at 3.21% per year over 2005–2026.

The two equity assumptions bracket the historical outcome. Over a five-year hold, the horizon of the headline result, buying needs 2.27% annual appreciation to match a portfolio compounding at the cautious 5.9%, comfortably below the 3.21% the dwelling actually delivered. Against a portfolio compounding at the realised 10.4%, it needs 3.24%, which is the realised rate almost exactly. Whether buying looks attractive today therefore depends on what one expects of global equities and UK house prices.

6.3 Regional break-even rates

The same calculation region by region, with each region's June 2026 price and rent, the national mortgage rate and region-specific stamp duty, gives a five-year requirement ranging from 1.17% in the North East to 2.90% in the South East under the cautious assumption, and from 2.04% to 3.94% under the realised one. Figure 7 maps the gradient.

The gradient tracks the price-to-rent ratio rather than the price level, with one exception. London has the third-lowest ratio of the nine at 20.1, but the second-*highest* break-even at 2.82%, because at £553,870 the average London dwelling sits above the £500,000 ceiling at which first-time-buyer stamp-duty relief is withdrawn altogether. That single threshold costs a London first-time buyer £17,694 against nothing at all in five of the nine regions, which moves the region three places up the ranking. What makes London expensive to enter is the stamp duty rather than the valuation.

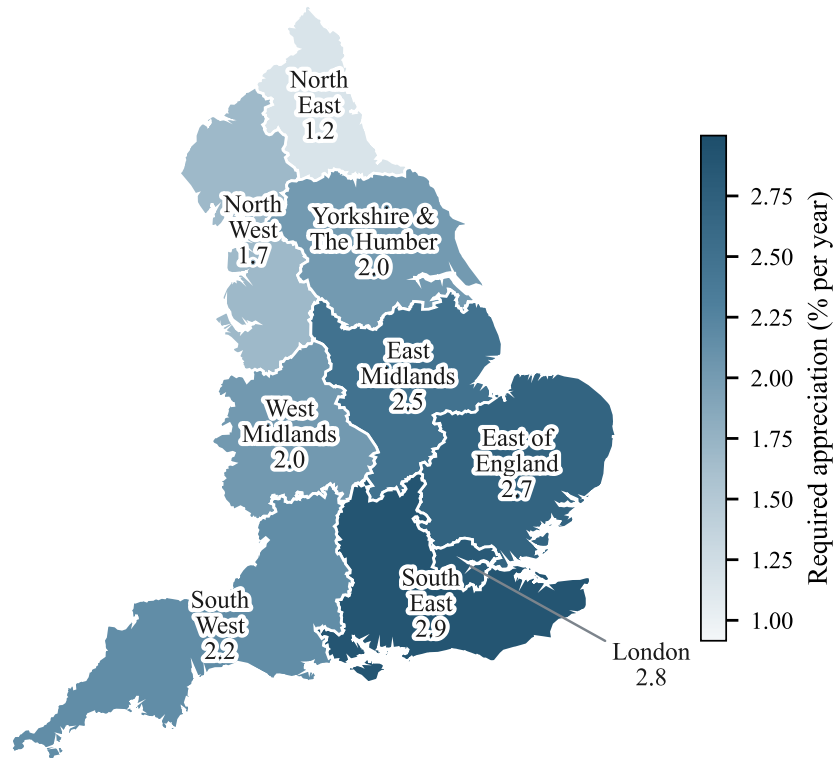


Figure 7: Forward-looking five-year required appreciation rate against global equities at the cautious 5.9% assumption, by English region, at June 2026 conditions (per cent per year). Darker shading indicates a higher break-even, that is, a market in which buying is harder to justify.

7 Robustness checks

Every value in the baseline of Table 2 is typical rather than universal. Tables 5 and 6 vary each parameter in turn, reporting the January 2005 full-period ratio, the median ratio across the 199 fixed five-year cohorts, and the share of those cohorts won by buying.

Table 5: Sensitivity to maintenance and deposit

Parameter	R (2005)	Median R	Buy-win %
<i>Maintenance and depreciation, UK-grounded range (% of value p.a.)</i>			
1.2	1.24	0.92	63
1.5 (baseline)	1.44	0.98	51
1.8	1.65	1.04	46
<i>Deposit (% of purchase price)</i>			
5	1.42	1.03	46
10 (baseline)	1.44	0.98	51
20	1.59	0.97	54
50	2.00	1.11	20
100 (cash)	2.59	1.22	8

Notes: $R > 1$: renting-and-investing produced greater terminal wealth. All other parameters at baseline.

Table 6: Sensitivity to other parameters

Parameter	<i>R</i> (2005)	Median <i>R</i>	Buy-win %
<i>Mortgage term</i>			
25 years	1.56	1.00	51
30 years (baseline)	1.44	0.98	51
35 years	1.36	0.97	53
<i>Rate fixation</i>			
2-year fixes	1.40	0.87	65
5-year fixes (baseline)	1.44	0.98	51
10-year fixes	1.46	0.98	51
<i>Renter's leverage</i>			
1.00× (baseline, unlevered)	1.44	0.98	51
1.50×	1.69	1.07	45
2.00×	1.93	1.15	38
<i>Transaction costs</i>			
Purchase costs 1.0%	1.43	0.97	54
Purchase costs 2.0%	1.49	1.02	49
Selling costs 1.0%	1.43	0.96	57
Selling costs 4.0%	1.49	1.07	46
Stamp duty excluded	1.39	0.95	57
<i>Renter's portfolio</i>			
MSCI ACWI net, less fee (baseline)	1.44	0.98	51
FTSE 100 total return	0.77	0.76	64

Notes: As Table 5. The renter-leverage rows charge margin interest at Bank Rate plus 1.50% and impose a 30% maintenance margin with forced liquidation; no cohort is liquidated.

7.1 Maintenance moves the answer most

Across the range UK national-accounts evidence supports, 1.2% to 1.8% of value per year, the buy-win share moves from 63% to 46% and the median ratio from 0.92 to 1.04. The verdict shifts from “buying modestly ahead” to “renting modestly ahead” inside the defensible band, so the near-parity headline is conditional on this calibration.

7.2 Leverage helps the buyer up to about 15 per cent down

An unlevered buyer is comprehensively beaten: at a 100% deposit, buying wins 8% of cohorts and the January 2005 ratio is 2.59. At a 50% deposit it wins 20%. Leverage is what allows a claim on an asset appreciating at 3.2% per year to compete with a portfolio compounding at 10.4%, and the result echoes the portfolio-choice observation that owner-occupied housing is attractive largely through its financing (Flavin and Yamashita, 2002).

But the relationship is not monotonic. Buying does *best* at deposits of 15–20% (54% of cohorts), slightly worse at the 10% baseline (51%), and worse again at 5% (46%). Leverage magnifies losses more than gains, and the cohorts that bought into the 2007–2008 peak were destroyed in proportion to how levered they were. A 5% deposit produces both the best and the worst outcomes, and the median cohort is worse off for it. The first-time buyer's high loan-to-value both makes ownership competitive and makes badly-timed ownership catastrophic.

7.3 A home-biased renter loses

Substituting the FTSE 100 for the global portfolio moves the buy-win share from 51% to 64% and the January 2005 ratio from 1.44 to 0.77, turning a renter who finished 44% ahead into one who finished 23% behind. The three-point annual gap between UK large-cap and global equity returns (7.1% against 10.4%) compounds into the difference between the two verdicts. The renter's side of the parity is earned by disciplined, globally diversified investing, not by renting as such; a home-biased renter loses.

Letting the renter borrow to invest works in the expected direction: at $2.00\times$ leverage, charging margin interest at Bank Rate plus 1.50%, the buy-win share falls to 38%. Two caveats belong with that figure. No cohort is force-liquidated even at $2\times$, but that is a property of *this* design rather than of leverage. Because the renter is cash-flow matched, the contributions made over sixty months exceed the entire initial stake: £32,764 against £24,314 for the November 2007 cohort, the worst entry point in the sample. The position therefore deleverages itself well before a margin call can bite. The minimum equity ratio across all cohorts is 31.9%, uncomfortably close to the 30% maintenance margin, and monthly observation understates drawdowns. A daily simulation would very likely show some forced sales. Separately, $2\times$ survived 2020 only because sterling fell about 13% in a global equity crash, which is a property of the currency pair in that episode and not of equity leverage.

7.4 Transaction costs, stamp duty and the term are second-order

None of these moves the buy-win share by more than a handful of percentage points. Doubling purchase costs from 1.0% to 2.0% costs five points; varying selling costs from 1.0% to 4.0% costs eleven; removing stamp duty entirely gains six. England's relatively low round-trip transaction costs are a structural advantage of ownership, and the period-accurate stamp-duty module, while conceptually important, changes little at average first-time-buyer prices, though not in London, where the relief cliff above £500,000 is worth £17,694. The mortgage term is almost inert: 25, 30 and 35 years give 51%, 51% and 53%.

8 Discussion

8.1 International context

Table 7 places English house-price growth in international context using BIS residential property price statistics (Bank for International Settlements, 2026). The United Kingdom sits mid-table over 2005–2025: cumulative nominal growth of 95% (3.3% per year).

Table 7: Nominal house-price growth, selected countries, 2005Q1–2025Q4

Country	Cumulative	Annualised
Hungary	356%	7.6%
Portugal	156%	4.6%
Netherlands	110%	3.6%
United Kingdom	95%	3.3%
United States	89%	3.1%
Germany	85%	3.0%
Ireland	62%	2.3%
France	60%	2.3%
Spain	60%	2.3%
Italy	15%	0.7%

Source: BIS residential property prices, nominal, selected series. The UK entry covers the whole United Kingdom; the representative English series of this paper grew 96% (3.2% per year) over the same span.

8.2 Limitations

Several limitations qualify the results. *Aggregation.* The representative dwelling is a construct; households decide in a specific market, and the regional results of Section 5.3, with buy-win shares from 38% to 63%, may differ from the aggregate result. Even within regions, composition differences between the owned and rented stocks mean the price-to-rent *level* carries measurement error. *Behavioural assumptions.* The renter invests 100% of every monthly difference, immediately and for two decades, in a globally diversified portfolio. The saving and portfolio-behaviour literature (Laibson, 1997; Thaler and Benartzi, 2004) puts this well above typical conduct; the FTSE-only robustness check shows how costly even a diversification failure is, before any failure of discipline. *Taxes.* The symmetric no-tax treatment is appropriate for a typical first-time buyer (no capital-gains tax on a main residence; ISA capacity covering the renter’s flows for most of the sample) but understates the drag on very large taxable portfolios. *Inference.* The 199 cohorts overlap and share the sample’s single realisation of asset returns, so cohort counts are counts of simulations rather than of independent observations. Win shares and correlations describe this history and are not tests on independent draws. *Unpriced attributes.* Ownership insures against rent risk (Sinai and Souleles, 2005) and carries control rights; renting

offers more flexibility and mobility (Ministry of Housing, Communities and Local Government, 2025). None of these enters the terminal-wealth accounting.

9 Conclusion

This paper compares buying a first home with renting and investing for every monthly entry point available in the English data, using matched housing consumption and monthly cash flows. The central result is that there is no general financial winner. Across 199 five-year cohorts, buying produces greater terminal wealth in 102 (51%), with a median renter-to-owner wealth ratio of 0.98 and a median gap of just £994 in the buyer's favour. The conventional view of renting as financially inferior is therefore difficult to reconcile with the historical evidence: for a renter who consistently invests the capital not committed to homeownership, the two strategies have been remarkably competitive.

What matters most is when the household buys, how long it stays, and how the purchase is financed. Buying performs better at longer horizons, winning 62% of seven-year cohorts and 61% of ten-year cohorts, although median renter wealth remains within 8–11% of owner wealth. At short horizons, transaction costs dominate: buying wins only 30% of one-year cohorts, and the round-trip costs of ownership take roughly six years to amortise. Entry conditions also matter substantially. Housing valuation and mortgage rates at purchase help determine which strategy wins, while regional five-year buy-win rates range from 38% in the North East to 63% in the West Midlands. These patterns should not, however, be treated as universal properties of tenure: they emerge from a sample containing a housing crash, a prolonged period of unusually cheap credit, and an exceptional global equity bull market.

Leverage is central to why buying remains competitive despite house prices appreciating much more slowly than global equities over the sample. An unlevered cash buyer wins only 8% of cohorts. But leverage also amplifies poor timing, and greater leverage is not monotonically better: buying performs best at deposits of 15–20%, while highly leveraged purchases are more exposed to house-price declines and selling costs. Conversely, the renter's performance is not automatic. It requires the discipline to invest the deposit, purchase costs, and subsequent monthly savings in a diversified portfolio rather than consume them.

The historical comparison can therefore inform the decision, but cannot settle it for a household buying today. For that purpose, the required appreciation rate provides a more useful benchmark. Under June 2026 conditions, a five-year buyer requires annual house-price appreciation of 2.3% to match a renter earning a cautious 5.9% equity return, and 3.2% to match the 10.4% equity return realised over the historical sample. Plausible assumptions therefore continue to place the two strategies on either side of the break-even point.

The broader conclusion is simple: homeownership is not inherently the financially superior way to pay for housing, nor does renting represent money simply “lost”. Buying combines leveraged exposure to housing with substantial transaction and user costs; renting preserves cap-

ital that can instead be invested elsewhere. Once both sides of that trade-off are accounted for consistently, the financial choice between them is much closer than conventional wisdom suggests. It depends less on tenure itself than on valuation, financing conditions, holding period, investment returns, and household behaviour.

All data, code, and an interactive implementation of the framework are openly available, allowing the comparison to be updated as new official data become available.

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Appendices

A Data construction

Rents

The rent series splices two ONS products. The PIPR historical series (released March 2025) provides average monthly rents in GBP for England and the nine English regions from January 2005 to February 2025, chain-linked to the predecessor Index of Private Housing Rental Prices; the live PIPR monthly dataset provides the same measures from January 2015 onwards. Over their 122-month overlap (January 2015–February 2025) the two agree exactly at every observation, so the splice, historical values before January 2015 and live values thereafter, introduces no discontinuity. Rent levels are model-consistent estimates published by the ONS (base-period mean rents uprated by the index), not raw sample means.

House prices

Regional and national average prices (all dwellings) are taken directly from the UK HPI full file (June 2026 vintage). The UK HPI revises its most recent months; the June 2026 observation used here is the first published estimate.

Representative dwelling

The representative price and rent are fixed-weight averages of the nine regional series with mid-2023 ONS population weights (millions): North East 2.65, North West 7.52, Yorkshire and The Humber 5.56, East Midlands 5.02, West Midlands 6.11, East of England 6.40, London 8.87, South East 9.46, South West 5.80. Using private-rented-sector household counts instead of population moves the June 2026 representative price-to-rent ratio from 20.7 to 20.6, and no headline result materially.

Mortgage rate

The baseline rate is a five-year fix at 90% loan-to-value, refixed every sixty months and repriced at each refix by the borrower's current loan-to-value (Section 3.3). No single Bank of England series covers it, so it is constructed from the quoted-rate family (Bank of England, 2026), using five-year data wherever possible so that no assumption about the term structure is needed:

- **February 2019 onwards:** the published five-year 90% LTV series (IUMZO28) directly.
- **Before February 2019:** interpolated in loan-to-value between the published five-year series at 75% (IUMBV42) and 95% (IUM5WTL), weight 0.55. The weight is fitted against IUMZO28 over the ninety months where it exists; mean discrepancy 0.01 percentage points, standard deviation 0.24. It is below the 0.75 that linear interpolation implies because the premium is convex in loan-to-value.

- **October 2008 to October 2013:** IUM5WTL is unpublished for all sixty-one months, because fewer than three lenders offered a 95% LTV five-year fix and the Bank suppresses averages below that threshold. The window is filled from the two-year 90% series (IUMB482) plus 0.30 percentage points, calibrated against contemporaneous quoted products; it puts January 2010 at 6.91% against a market range of 6.49% to 7.49%. IUMB482 is itself unpublished for March to May 2009 and is interpolated across those three months. The patch is not level-shifted to meet the interpolation, so there is a step of about 0.3 percentage points at each seam.
- **Bands.** The 75% and 95% series are carried through so the engine can reprice by current loan-to-value, piecewise-linearly between bands, with the 75% rate used unchanged below 75%.

The resulting series averages 4.74% and ranges from 2.33% to 6.99%, sitting 1.19 percentage points above CFMBJ95.

Five-year data is used in preference to transplanting a two-year loan-to-value premium onto a five-year rate, because the loan-to-value premium compresses at longer tenors when high-loan-to-value credit is rationed: the Bank's data give a five-year-minus-two-year term premium of 0.45 percentage points at 75% loan-to-value against 0.27 at 95%. A transplant is unbiased after 2019 but puts January 2010 at 8.20%, above anything then available.

CFMBJ95 (Bankstats table G1.4, the weighted-average rate on new advances to households, January 2004 onwards) is retained for the floating-rate case in Section 7. Bank Rate (IUMABEDR) supplies the margin cost in the renter-leverage case.

Equity returns

MSCI publishes end-of-month ACWI index levels directly in GBP. I use the net-total-return variant (dividends reinvested after withholding tax) and convert levels to simple monthly returns. The baseline series deducts a fund fee of 0.12% per year, applied as a constant monthly drag: $r_t^{\text{net}} = (1 + r_t)(1 - 0.0012)^{1/12} - 1$. The FTSE 100 total-return comparator is the net-asset-value total return of a low-cost FTSE 100 ETF (dividends reinvested at NAV; ongoing charge 0.40% per year to June 2014, 0.07% thereafter), which tracks the FTSE 100 total-return index with correlation 0.9999 over the months both are available.

Replication

All raw data under the Open Government Licence, all cleaning and analysis code, and scripts that re-download the licence-restricted financial series from their public endpoints are available at <https://github.com/Martin-Cimprich/buy-vs-rent-england>.

B Stamp duty schedules

Table 8 summarises the SDLT liability of a first-time buyer in England by purchase month, as implemented in the simulation. Before 4 December 2014 SDLT was a “slab” tax (the rate applied to the whole price); thereafter a “slice” tax (marginal bands). Where first-time-buyer relief exists, the relief schedule applies below its price cap and the standard schedule above it.

Table 8: SDLT for an England first-time buyer, by period

Period	Schedule applying to a first-time buyer
to 16 Mar 2005	Slab: 0% to £60k; 1% to £250k; 3% to £500k; 4% above
17 Mar 2005–22 Mar 2006	Slab, nil-rate band £120k
23 Mar 2006–2 Sep 2008	Slab, nil-rate band £125k
3 Sep 2008–31 Dec 2009	Slab, nil-rate band £175k (holiday)
1 Jan 2010–24 Mar 2010	Slab, nil-rate band £125k
25 Mar 2010–24 Mar 2012	First-time-buyer relief: 0% to £250k; standard slab above
25 Mar 2012–3 Dec 2014	Slab, nil-rate band £125k (relief withdrawn; 7% top band from Mar 2012)
4 Dec 2014–21 Nov 2017	Slice: 0% to £125k; 2% to £250k; 5% to £925k; 10% to £1.5m; 12% above
22 Nov 2017–7 Jul 2020	FTB relief: 0% to £300k, 5% £300–500k (no relief above £500k)
8 Jul 2020–30 Jun 2021	Holiday: 0% to £500k for all buyers; slice above
1 Jul 2021–30 Sep 2021	Standard nil-rate £250k; FTB relief 0% to £300k, 5% to £500k
1 Oct 2021–22 Sep 2022	Standard slice (£125k nil); FTB relief 0% to £300k, 5% to £500k
23 Sep 2022–31 Mar 2025	FTB relief: 0% to £425k, 5% £425–625k (no relief above £625k)
1 Apr 2025–	FTB relief: 0% to £300k, 5% £300–500k (no relief above £500k)

Sources: HMRC SDLT rate tables 2003–2026; House of Commons Library briefings SN07050 and CBP-9814. The simulation evaluates the schedule at each month’s representative (or regional) average price. Scotland (LBTT, from April 2015) and Wales (LTT, from April 2018) operate different schedules and are outside the scope of this paper.